

ASL RECOGNITION AND TRANSLATION PROJECT

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ABSTRACT

This project presents a real-time American Sign Language (ASL) recognition system using deep learning models and webcam-based inference. Three models were developed and evaluated: a simple convolutional neural network (CNN), a residual-based ResNet-18 model with frozen layers, and a fine-tuned ResNet-18 model with data augmentation. Experimental results show that the fine-tuned ResNet-18 achieved near-perfect classification accuracy of 99.99% across 29 ASL classes. The system was integrated into a real-time pipeline using MediaPipe for hand detection, OpenCV for video capture, and text-to-speech (TTS) for auditory feedback. This demonstrates the feasibility of deploying deep learning-based ASL recognition systems in real-time environments. Limitations and future directions for continuous sign recognition are discussed.

Index Terms— American Sign Language, Deep Learning, Computer Vision, ResNet-18, Real-Time Systems, Model Comparison, Augmentation

GITHUB REPOSITORY

<https://github.com/Cacifyy/ASL-Translation-and-Text-to-Speech.git>

1. INTRODUCTION

American Sign Language (ASL) is a primary means of communication for many individuals in the deaf and hard-of-hearing community, covering approximately 3% of the population in North America [4]. However, communication barriers still exist when interacting with individuals who do not understand ASL. Automated ASL recognition systems can help bridge this gap by translating gestures into text or speech in real time.

Recent advancements in deep learning, particularly convolutional neural networks (CNNs), have significantly improved performance in image classification tasks [3]. Residual architectures such as ResNet have further enabled deeper and more effective models [1]. These methods can be applied to ASL recognition, where hand gestures must be classified accurately under varying conditions.

In this project, we develop and compare multiple deep learning models for ASL classification and integrate the best-performing model into a real-time inference system. The main contributions of this work include:

- Implementation and comparison of three ASL classification models
- Exploration of residual architectures and fine-tuning strategies for improved performance
- Development of a real-time webcam-based ASL recognition system with text-to-speech output

2. RELATED WORK

Deep learning approaches have become the dominant method for ASL recognition, particularly using convolutional neural networks trained on image datasets [2]. CNN-based models have demonstrated strong performance in recognizing static hand gestures due to their ability to extract spatial features from images [3].

Residual networks (ResNet) introduced skip connections that allow deeper architectures to be trained effectively without degradation [1]. These architectures have been widely adopted for image classification tasks due to their strong performance and stability.

Real-time ASL recognition systems introduce additional challenges, including latency, hand detection accuracy, and environmental variability. Many systems incorporate computer vision pipelines such as MediaPipe for real-time hand tracking and region-of-interest extraction [8].

Data augmentation techniques are commonly used to improve generalization by increasing dataset variability and robustness to real-world conditions [7].

3. MATERIALS AND METHODS

3.1 Dataset

The dataset used in this project is the ASL Alphabet Dataset obtained from Kaggle [5]. This dataset consists of a large collection of labeled images representing hand signs corresponding to the American Sign Language (ASL) alphabet. The dataset is organized into 29 distinct classes, including the 26 letters (A–Z) and three additional classes: space, delete, and nothing. These additional classes are particularly important for real-time applications, as they allow the model to handle transitions, corrections, and non-sign states.

The training dataset contains approximately 87,000 images, each with a resolution of 200×200 pixels. The images are evenly distributed across the classes, providing a balanced dataset for training deep learning models. In

contrast, the test dataset contains only 29 images (one per class), which encourages the use of external or real-world data for evaluation rather than relying solely on the provided test set.

3.2 Preprocessing

All images were preprocessed using a standardized pipeline to ensure consistency between training and inference. Each image was:

- Resized to 224×224 pixels
- Converted from BGR to RGB format
- Normalized using ImageNet mean and standard deviation

This preprocessing ensures compatibility with pretrained convolutional architectures.

3.3 Model Architectures

3.3.1 Simple CNN

A baseline convolutional neural network was implemented based on standard image classification tutorials. This model consists of stacked convolutional and pooling layers followed by fully connected layers.

CNNs are well-suited for image classification tasks due to their ability to automatically learn hierarchical spatial features [3]. This model was chosen as a simple and interpretable baseline to evaluate how well a custom architecture could perform on the ASL dataset.

3.3.2 ResNet-18 (Frozen)

A ResNet-18 architecture was implemented with its convolutional layers frozen, meaning that their weights were not updated during training. Only the final fully connected layer was trained for the ASL classification task.

Freezing layers reduces the number of trainable parameters and allows faster training, while testing whether general visual features learned by deep convolutional layers are sufficient for the task. Residual networks such as ResNet-18 are designed to learn deep hierarchical representations using skip connections, which improve training stability [1].

This model was used to evaluate whether fixed feature representations from a deep architecture could generalize to ASL gesture recognition without full retraining.

3.3.3 ResNet-18 (Fine-Tuned with Augmentation)

The final model builds on the ResNet-18 architecture by unfreezing deeper layers and allowing them to update during training. This process enables the model to adapt its learned features specifically to the ASL dataset.

In addition, data augmentation techniques were applied during training to increase dataset variability and improve generalization performance [7]. These techniques help the

model become more robust to variations in lighting, hand orientation, and background conditions.

This model was selected to evaluate whether adapting deeper layers of a high-capacity architecture would improve classification accuracy compared to fixed feature extraction.

3.3.4 Model Implementation

The model implementations were adapted from prior coursework and tutorials, where similar architectures were applied to image classification tasks. These implementations were refactored and extended for the ASL recognition task by modifying input pipelines, adjusting output classes, and integrating augmentation and fine-tuning strategies. This approach ensured a structured experimental setup while tailoring the models to the specific dataset. The final architecture consists of a ResNet-18 backbone with a modified classification head for 29 classes.

3.4 Training Setup

All models were trained using cross-entropy loss and the Adam optimizer with a learning rate of 0.001. Training was performed using mini-batches (batch size = 32) over 10 epochs, which provided a balance between convergence and training time.

Model performance was evaluated using validation accuracy and F1-score, and the best-performing weights were saved based on validation results to reduce overfitting. Each model outputs predictions over 29 classes. Data augmentation (e.g., random transformations) was applied only in the fine-tuned experiment to improve generalization to variations in hand position and lighting.

All training and inference were implemented in PyTorch, with automatic device selection between CPU and GPU.

3.5 Real-Time System Pipeline

A real-time ASL recognition system was developed using a modular pipeline:

1. **Webcam Input:** Frames are captured using OpenCV-based utilities
2. **Hand Detection:** MediaPipe is used to detect hand landmarks and isolate the region of interest [8]
3. **Hand Cropping:** A bounding box is extracted around the detected hand
4. **Model Inference:** The cropped image is passed through the trained model to obtain predictions
5. **Output Display:** Predictions are shown visually and optionally spoken using text-to-speech

The system runs continuously and updates predictions in real time using a webcam interface.

4. RESULTS AND DISCUSSION

4.1 Model Performance

Figure 1 shows the training and validation loss and accuracy for each model. All models converge quickly, with performance stabilizing after approximately 6 epochs. The similar trends between training and validation indicate minimal overfitting, hence our epoch limit of 10.

The performance of all three models is summarized in Table 1. The fine-tuned ResNet-18 achieved the highest accuracy (99.99%), outperforming the simple CNN (99.38%) and frozen ResNet-18 (88.39%).

The frozen ResNet-18 showed significantly lower performance, suggesting that fixed feature representations are insufficient for ASL gesture recognition. In contrast, fine-tuning deeper layers enables the model to better capture task-specific features.

Confusion matrices (Figures 2–4) support these results, showing widespread errors in the frozen model, minor confusion in the CNN, and near-perfect predictions in the fine-tuned model.

Table 1. Comparison of model performance

Model	Accuracy	Macro F1
Simple CNN	99.38%	0.9939
ResNet-18 (Frozen)	88.39%	0.8843
ResNet-18 (Fine-Tuned)	99.99%	0.9999

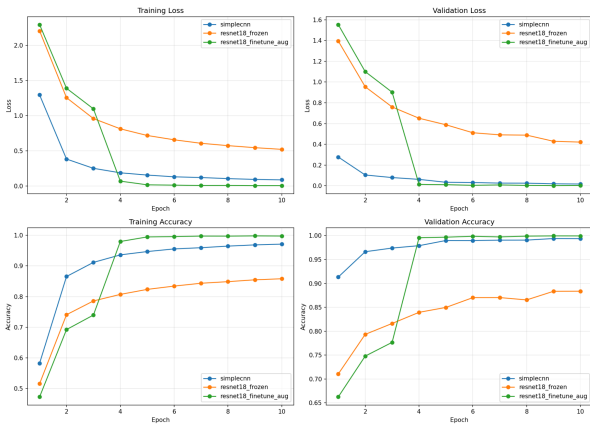


Fig. 1. Validation and Testing Results for Loss and Accuracy



Fig. 2. CNN Confusion Matrix (heatmap did not produce enough variability for errors, thus number matrix was used)

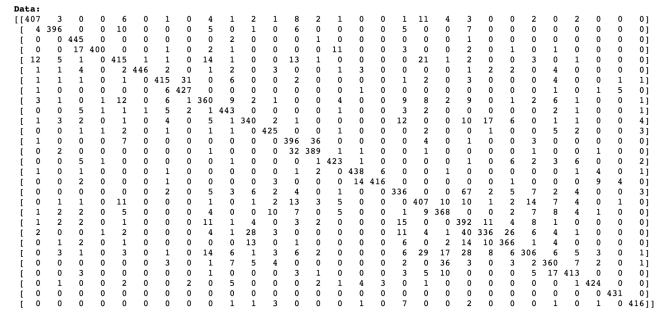


Fig. 3. ResNet-18 (Frozen) Confusion Matrix

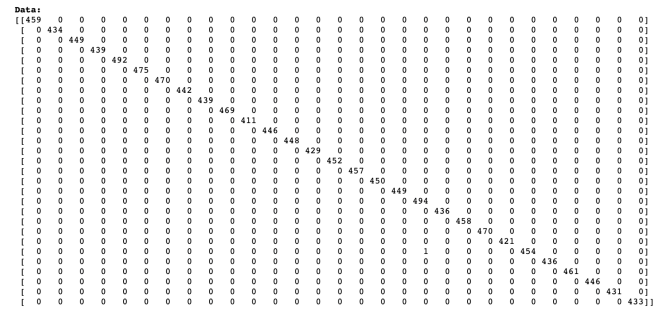


Fig. 4. ResNet-18 (Fine Tuned) Confusion Matrix

4.2 Class Level Analysis

A detailed analysis of class-level performance was conducted using classification reports for each model. The frozen ResNet-18 model exhibited significantly lower precision and recall for several classes, particularly for letters such as U, V, X, and R, which achieved F1 scores below 0.80 in some cases. These errors were concentrated among visually similar gestures.

The CNN achieved consistently high performance across classes, but the fine-tuned ResNet-18 achieved almost perfect precision and recall. This indicates that deeper models, when trained properly, learn more discriminative features.

4.3 Misclassification Analysis

To better understand model limitations, misclassified samples were analyzed. For the simple CNN and frozen ResNet-18 models, errors were primarily observed between visually similar hand gestures.

For example, one misclassification involved the letter X being predicted as S, indicating that the model struggled to distinguish subtle differences in finger positioning.

These types of errors highlight the challenges of distinguishing fine-grained hand configurations in static images, particularly when gestures share similar shapes.

The fine-tuned ResNet-18 model showed significantly fewer misclassifications, suggesting that adapting deeper layers enables the model to capture more detailed and discriminative features.

Overall, misclassification analysis reinforces the importance of model capacity and feature learning in achieving high-performance ASL recognition.

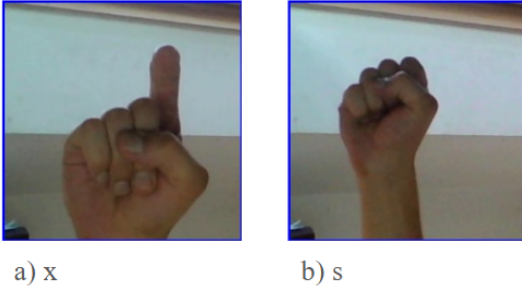


Fig. 5. ASL letters with similar spatial features

4.4 Real-Time Performance

The real-time system successfully performs ASL recognition using webcam input. The use of hand detection and cropping improves prediction accuracy by focusing on relevant regions.

The strong classification performance of the fine-tuned model directly translates to improved real-time predictions. During live inference, each frame is processed using a preprocessing pipeline and passed through the trained model to obtain class probabilities via softmax.

The system demonstrates stable predictions and can output spoken labels with minimal latency.

The model outputs a predicted label and confidence score, which are then displayed and optionally spoken using text-to-speech, enabling an interactive ASL recognition system.

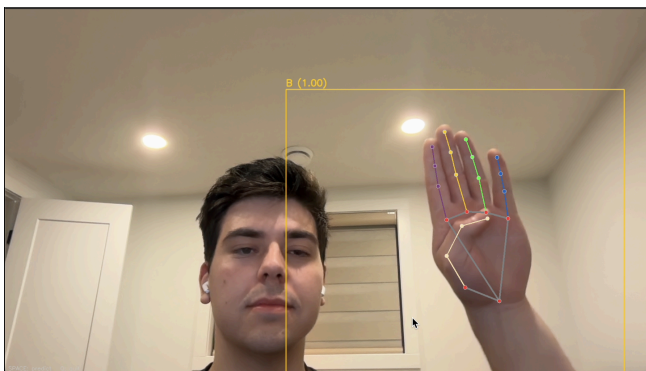


Fig. 6. Real Time ASL letter prediction for 'B' with a 1.00 confidence score.

5. LIMITATIONS

Despite strong performance, several limitations exist:

- The dataset consists of static images rather than continuous signing. The use case requires that there be a live video feed with real time translation. However, video feeds have different data markers that are not accounted for with pictures.
- Only single-hand gestures are considered
- No temporal modeling (e.g., video sequence analysis) is used.
- Performance may degrade under poor lighting or occlusion conditions
- Model trained on a controlled dataset and may not generalize perfectly to enough real-world environments.

6. FUTURE WORK

Several promising directions exist for extending this ASL recognition system. The most impactful next step would be incorporating temporal modeling, using architectures such as LSTMs or Transformer-based models to move beyond static image classification toward continuous sign language recognition from video sequences. This would allow the system to interpret multi-sign sentences rather than isolated letters, dramatically increasing real world utility. Additionally, the dataset could be expanded to include two-handed gestures, varied lighting conditions, and diverse skin tones and backgrounds to improve generalization. Finally, incorporating user feedback mechanisms could allow the system to adapt and improve over time through continual learning.

7. CONCLUSION

This project demonstrates the effectiveness of deep learning for ASL recognition through a comparison of multiple model architectures. A fine-tuned ResNet-18 model achieved near-perfect accuracy and was successfully integrated into a real-time webcam-based system.

The results highlight the importance of model architecture selection and fine-tuning strategies in achieving high performance, as well as the feasibility of deploying such models in real-time applications.

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